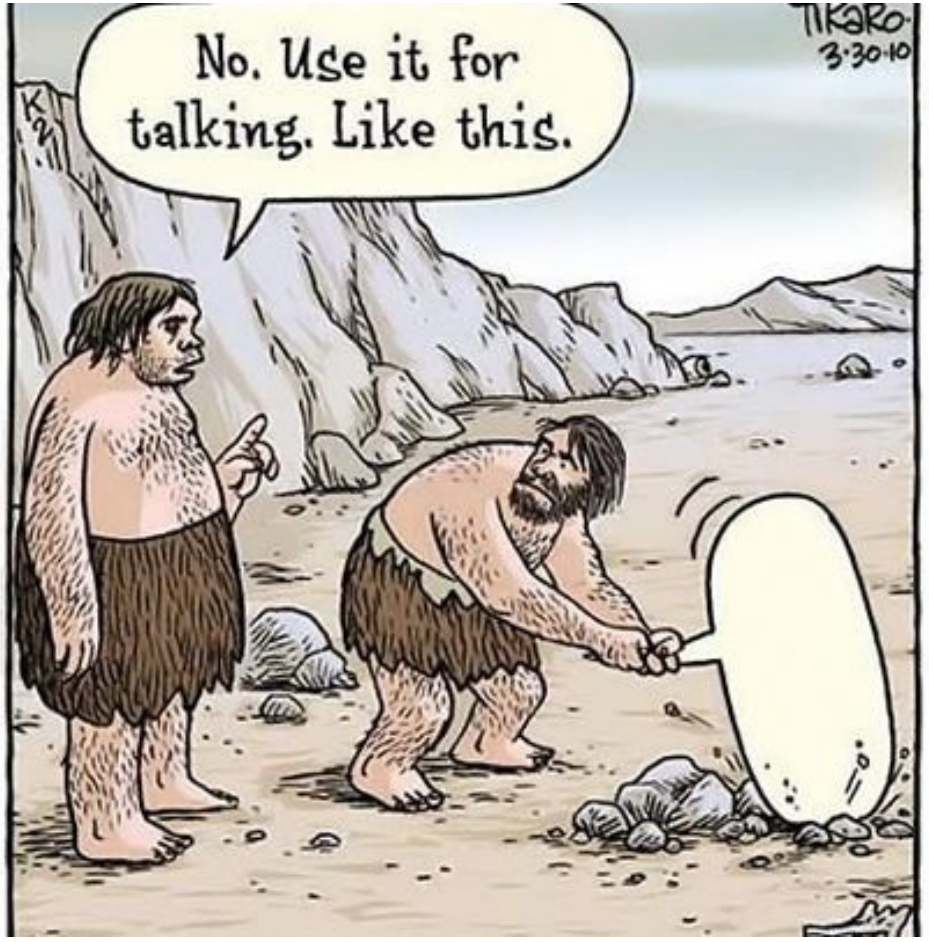
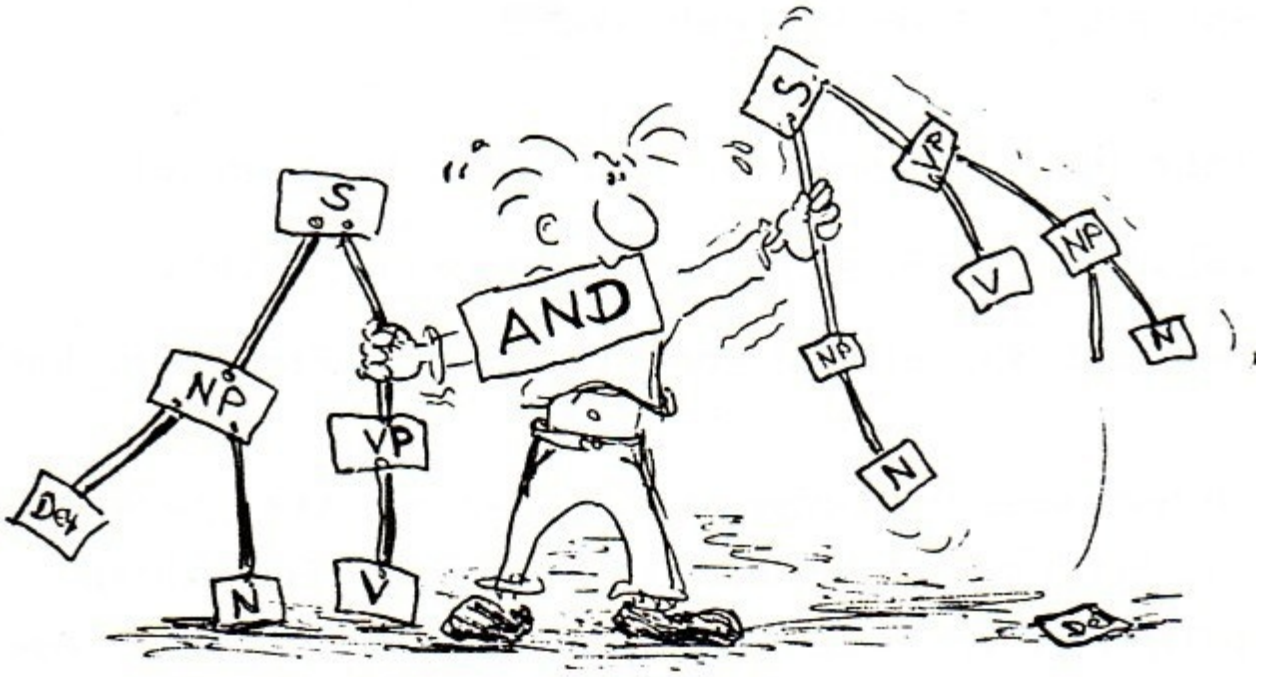
Natural Language Processing

Sentence Syntax



NLP: Syntax (Sentence Level, HMM)



Classical NLP >

Hidden Markov Model

- Used to model processes that move between states
- States are hidden, but we see state outputs
- Our goal: find the sequence of states

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Classical NLP > Hidden Markov Model

Example

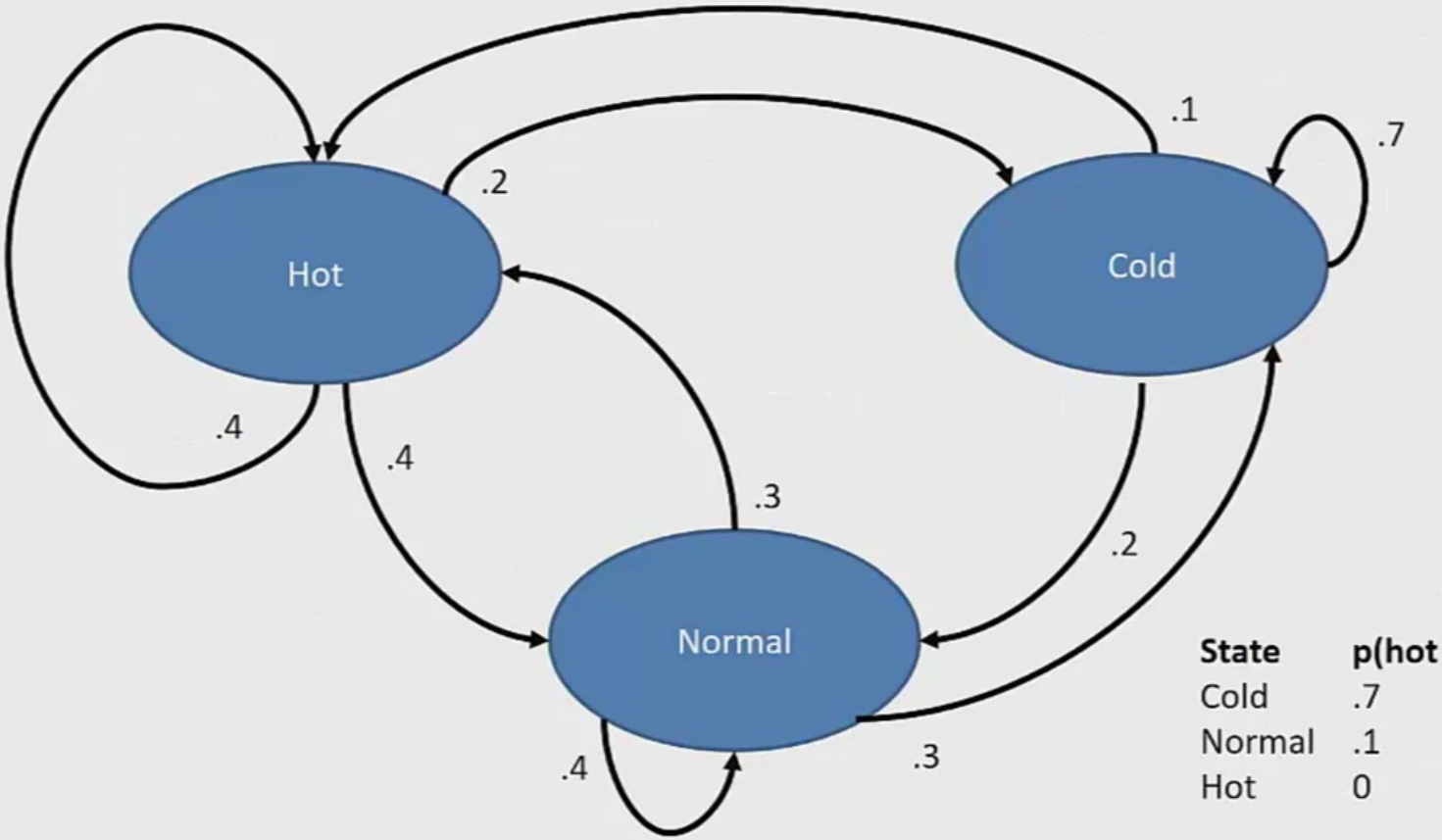
- Interested in the weather in a small town, week of Oct 5, 1900
- Weather records not kept, but we have diary of a teenager and his once a day snacks: hot chocolate, soda, or ice cream
- For our purposes, weather is: **cold**, **normal**, or **hot**
- If we are given:
  - probabilities of a given snack for a given weather day, e.g., p(soda | hot)
  - weather day transition probabilities, e.g., p(hot | cold)
  - diary: soda, soda, hot chocolate, ice cream, soda, soda, ice cream
- How can we find the weather sequence for the week?

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Classical NLP > Hidden Markov Model

Example – HMM Components



State	p(hot chocolate)	p(soda)	p(ice cream)
Cold	.7	.3	0
Normal	.1	.7	.2
Hot	0	.6	.4

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Classical NLP > Hidden Markov Model

Example – Viterbi Algorithm

Day 1: p(cell) = p(transition) *p(output)

$p(cell) = \max_{prev}[\, p(prev) * p(transition) \,] * p(output)$

State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
Cold	.7	.1	.2
Normal	.3	.4	.3
Hot	.2	.4	.4

State	p(hot chocolate)	p(soda)	p(ice cream)
Cold	.7	.3	0
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Classical NLP > Hidden Markov Model

Example – Viterbi Algorithm

Day 1: p(cell) = p(transition) *p(output)

$p(cell) = \max_{prev}[\ p(prev)*p(transition)\ ] *p(output)$

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Hot							
Normal							
Cold							
Diary:	**soda**	**soda**	**hot choc.**	**ice cream**	**soda**	**soda**	**ice cream**

State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
Cold	.7	.1	.2
Normal	.3	.4	.3
Hot	.2	.4	.4

State	p(hot chocolate)	p(soda)	p(ice cream)
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Example – Viterbi Algorithm

Day 1: p(cell) = p(transition) *p(output)

$p(cell) = \max_{prev}[\ p(prev)*p(transition)\ ] *p(output)$

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Diary:	**soda**	**soda**	**hot choc.**	**ice cream**	**soda**	**soda**	**ice cream**
Hot							
Normal							
Cold	.1 * .3 = .03						

State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
Cold	.7	.1	.2
Normal	.3	.4	.3
Hot	.2	.4	.4

State	p(hot chocolate)	p(soda)	p(ice cream)
Cold	.7	.3	0
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Classical NLP > Hidden Markov Model

Example – Viterbi Algorithm

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$p(cell) = \max_{prev}[\ p(prev)*p(transition)\ ] * p(output)$

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Diary:	**soda**	**soda**	**hot choc.**	**ice cream**	**soda**	**soda**	**ice cream**
Hot							
Normal	.8 * .7 = .56						
Cold	.1 * .3 = .03						

State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
Cold	.7	.1	.2
Normal	.3	.4	.3
Hot	.2	.4	.4

State	p(hot chocolate)	p(soda)	p(ice cream)
Cold	.7	.3	0
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Classical NLP > Hidden Markov Model

Example – Viterbi Algorithm

Day 1: p(cell) = p(transition) *p(output)

$p(cell) = \max_{prev}[\ p(prev)*p(transition)\ ] *p(output)$

Hot	.1 * .6 = .06						
Normal	.8 * .7 = .56						
Cold	.1 * .3 = .03						
	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Diary:	**soda**	**soda**	**hot choc.**	**ice cream**	**soda**	**soda**	**ice cream**

State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
Cold	.7	.1	.2
Normal	.3	.4	.3
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State	p(hot chocolate)	p(soda)	p(ice cream)
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Classical NLP > Hidden Markov Model

Example – Viterbi Algorithm

Day 1: p(cell) = p(transition) *p(output)

p(cell) = $\max_{prev}$[p(prev)*p(transition)] *p(output)

Hot	.1 * .6 = .06						
Normal	.8 * .7 = .56	Max(.03*.7, .56*.3, .06*.2)					
Cold	.1 * .3 = .03						
	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Diary:	**soda**	**soda**	**hot choc.**	**ice cream**	**soda**	**soda**	**ice cream**

State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
Cold	.7	.1	.2
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Classical NLP > Hidden Markov Model

Example – Viterbi Algorithm

Day 1: p(cell) = p(transition) *p(output)

p(cell) = $\max_{prev}$[p(prev)*p(transition)] *p(output)

Hot	.1 * .6 = .06						
Normal	.8 * .7 = .56	Max(.03*.7, .56*.3, .06*.2) = Max(.021, .168, .012)					
Cold	.1 * .3 = .03	.168 * .3 = .0504					
	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Diary:	**soda**	**soda**	**hot choc.**	**ice cream**	**soda**	**soda**	**ice cream**

State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
Cold	.7	.1	.2
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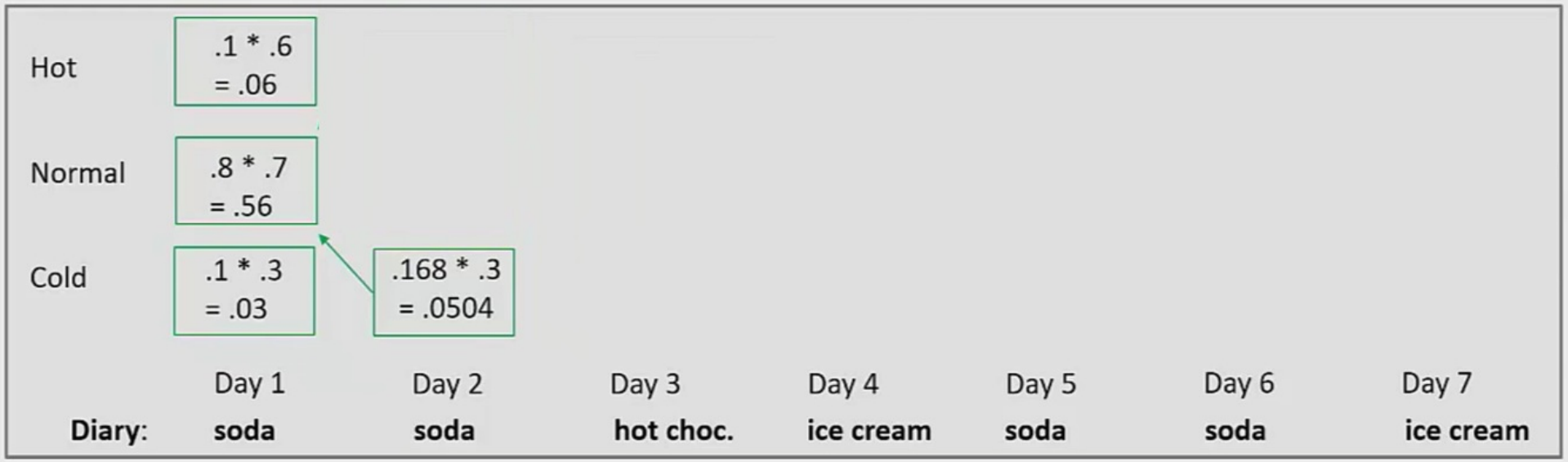
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Example – Viterbi Algorithm

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State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
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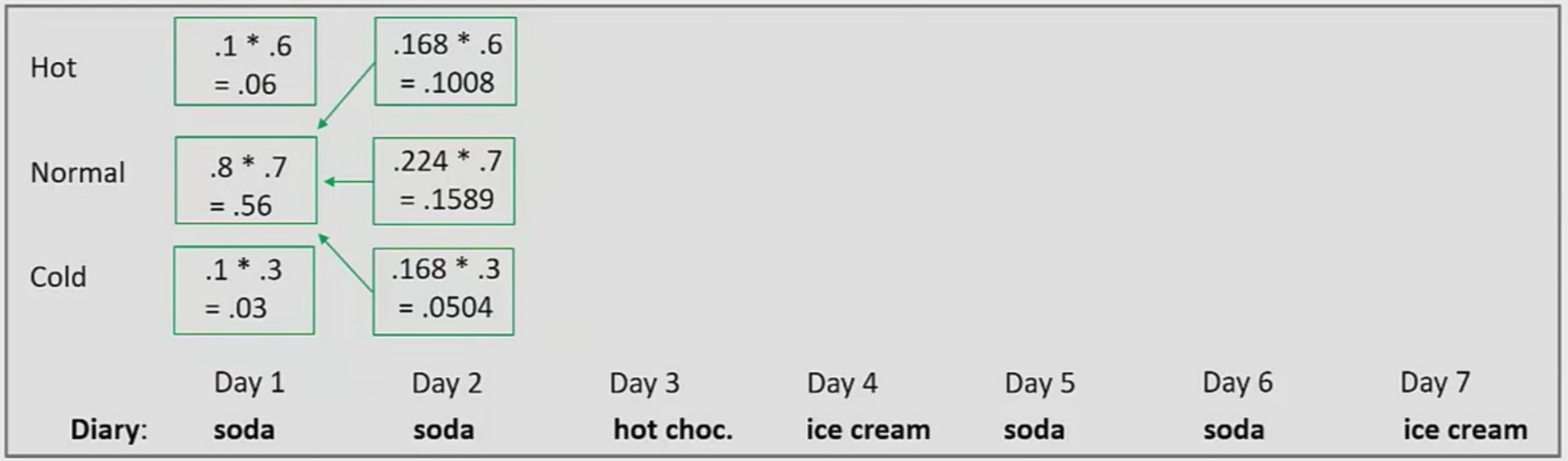
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Example – Viterbi Algorithm

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State	p(Cold)	p(Normal)	p(Hot)
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Microsoft

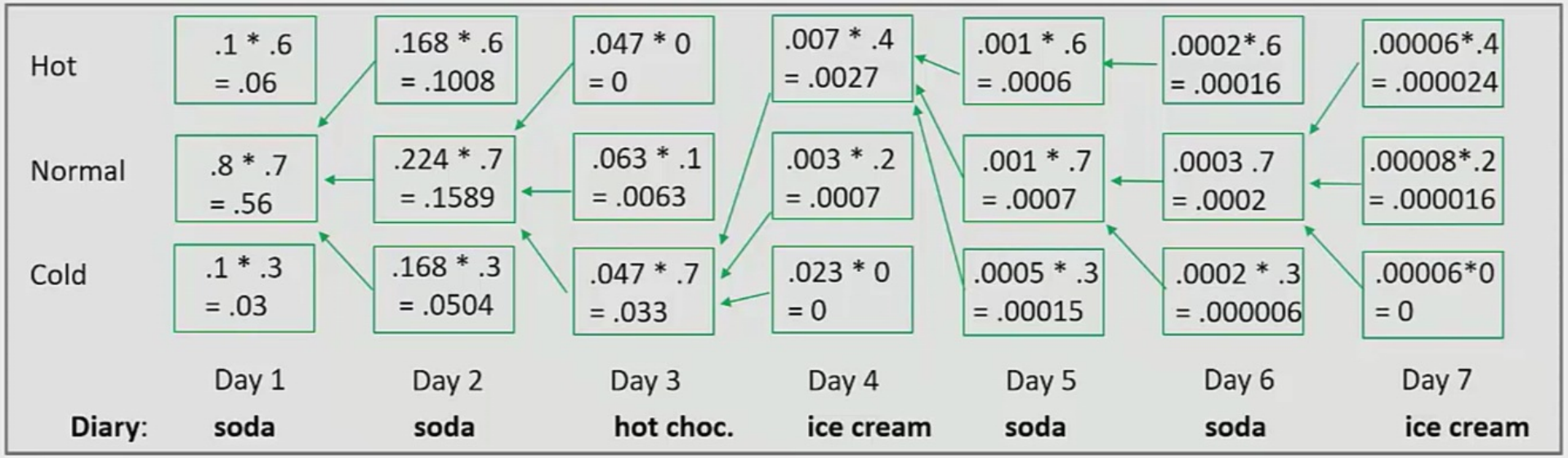
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Example – Viterbi Algorithm

Day 1: p(cell) = p(transition) *p(output)

p(cell) = $\max_{prev}$[p(prev)*p(transition)] *p(output)



State	p(Cold)	p(Normal)	p(Hot)
Start	.1	.8	.1
Cold	.7	.1	.2
Normal	.3	.4	.3
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State	p(hot chocolate)	p(soda)	p(ice cream)
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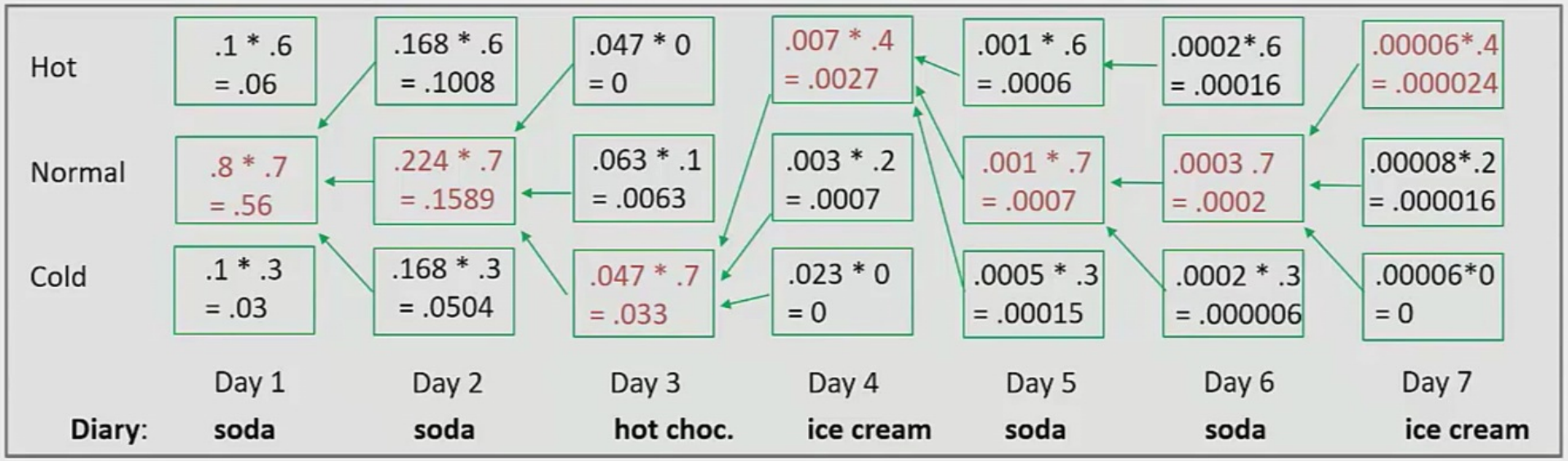
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Example - Viterbi algorithm

- The guessed sequence of states is

(' day=', 1, 'normal')
(' day=', 2, 'normal')
(' day=', 3, 'cold')
(' day=', 4, 'hot')
(' day=', 5, 'normal')
(' day=', 6, 'normal')
(' day=', 7, 'hot')

Example - Viterbi algorithm

- Playing with the Python code....

Maximum Entropy Markov Models (MEMM)

- While an HMM can achieve very high accuracy in POS tagging, it requires a number of architectural innovations to deal with unknown words, suffixes, and so on.
- Maximum Entropy Markov Models (MEMM) integrate the ideas of Hidden Markov Models and logistic regression, a probabilistic classifier that makes use of supervised machine.
- MEMM can be trained on POS-annotated training text for the language. This is the way the Stanford POS tagger works.
- Reference: https://nlp.stanford.edu/software/tagger.shtml

HMM Lab

- Experiment with the Hidden Markov Model python pieces of code available on the AulaWeb module.
- Extend the Hidden Markov Model python piece of code with other parts of speech (see for example https://en.wikipedia.org/wiki/Brown_Corpus#Part-of-speech_tags_used) and other words and transitions, and use it to POS-tag the "time flies like an arrow" sentence.
- Invent other short ambiguous sentences and extend the code in such a way to properly tag them.